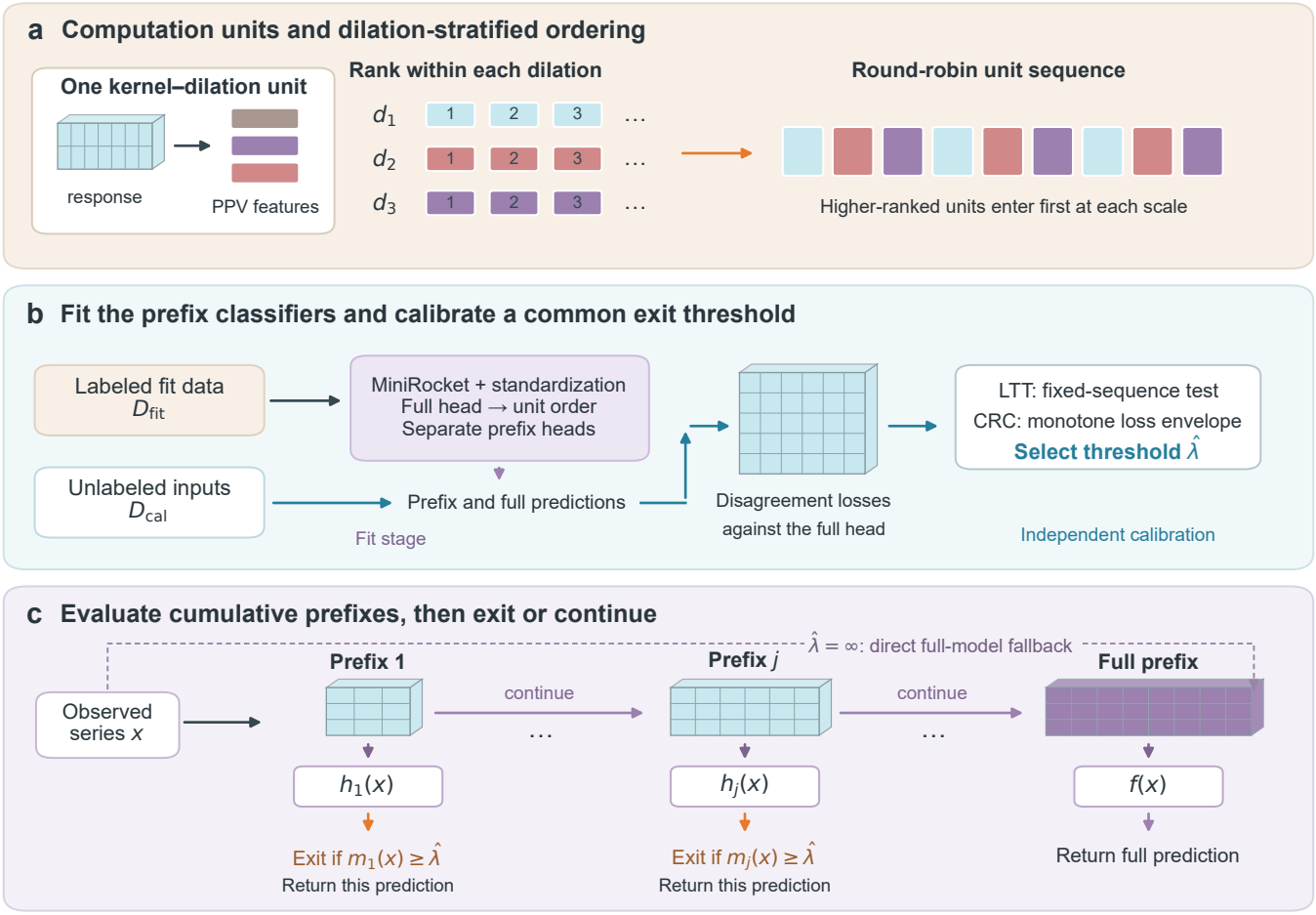

a Computation units and dilation-stratified ordering
One kernel–dilation unit
response
PPV features
Rank within each dilation
d_1 1 2 3 …
d_2 1 2 3 …
d_3 1 2 3 …
Round-robin unit sequence
Higher-ranked units enter first at each scale
b Fit the prefix classifiers and calibrate a common exit threshold
Labeled fit data D_{fit}
Unlabeled inputs D_{cal}
MiniRocket + standardization
Full head → unit order
Separate prefix heads
Prefix and full predictions
Fit stage
Disagreement losses against the full head
LTT: fixed-sequence test
CRC: monotone loss envelope
Select threshold $\hat{\lambda}$
Independent calibration
c Evaluate cumulative prefixes, then exit or continue
$\hat{\lambda} = \infty$: direct full-model fallback
Observed series x
Prefix 1
continue
…
Prefix j
continue
…
Full prefix
$h_1(x)$
$h_j(x)$
$f(x)$
Exit if $m_1(x) \geq \hat{\lambda}$
Return this prediction
Exit if $m_j(x) \geq \hat{\lambda}$
Return this prediction
Return full prediction